

JIZAN, Saudi Arabia

WMO: 411400

Lat: 16.8964N

Lon: 42.5837E

Elev: 6

StdP: 101.25

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.8	(c) 22.6	(d) 16.0	(e) 11.3	(f) 26.8	(g) 17.1	(h) 12.2	(i) 26.5	(j) 8.9	(k) 28.6	(l) 8.2	(m) 28.5	(n) 1.7	(o) 110	(p) 0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	38.3	27.9	38.0	28.1	37.2	28.3	30.5	36.0	30.1	35.7	29.7	35.3	5.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
29.1	25.8	34.8	28.8	25.3	34.6	28.2	24.4	34.4	102.3	35.5	100.3	35.8	98.9	35.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.1	(b) 8.0	(c) 7.2	(e) 19.4	(f) 40.5	(g) 1.5	(h) 1.1	(i) 18.3	(j) 41.3	(k) 17.4	(l) 42.0	(m) 16.5	(n) 42.6	(o) 15.4	(p) 43.4		
(a) 9.1	(b) 8.0	(c) 7.2	(e) 15.9	(f) 31.8	(g) 1.5	(h) 1.0	(i) 14.8	(j) 32.5	(k) 13.9	(l) 33.1	(m) 13.1	(n) 33.6	(o) 12.0	(p) 34.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.7	26.3	27.0	28.5	30.8	32.8	34.0	34.3	33.8	33.3	31.4	29.2	27.2
	DBStd	3.07	1.02	1.02	1.24	1.32	1.21	1.11	1.26	1.26	0.95	0.99	1.20	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7571	506	476	573	626	707	721	752	737	701	664	577	533
	CDD18.3	4529	247	242	314	376	448	471	493	478	451	406	327	274
	CDH23.3	65041	2372	2521	3872	5439	7033	7649	7966	7596	7118	6113	4347	3015
	CDH26.7	38019	634	762	1615	3069	4557	5249	5487	5119	4719	3652	2103	1051
Wind	WSAvg	3.2	3.0	3.0	3.1	3.1	3.1	3.3	3.9	3.6	3.1	2.9	2.9	2.9
Precipitation	PrecAvg	113	11	2	4	9	7	2	9	30	4	15	10	11
	PrecMax	275	61	15	26	67	49	13	105	120	35	151	61	80
	PrecMin	37	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	59	18	4	7	17	13	3	21	32	8	31	17	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	32.0	34.0	37.2	39.1	39.2	39.2	38.8	38.2	37.6	35.2	32.2
		MCWB	24.8	25.4	26.0	25.9	26.5	28.1	27.6	28.2	28.3	27.0	25.7	25.7
	2%	DB	30.7	31.1	33.0	36.0	37.9	38.1	38.1	37.9	37.8	36.4	34.2	31.8
		MCWB	24.7	25.1	26.0	26.6	27.2	28.3	27.9	28.5	28.6	27.6	26.2	25.5
	5%	DB	30.0	30.7	32.2	35.1	37.0	37.3	37.2	37.1	37.1	35.9	33.6	31.1
		MCWB	24.3	24.9	25.8	26.8	27.6	28.5	28.0	28.5	28.8	27.8	26.2	25.2
	10%	DB	29.3	30.0	31.8	34.2	36.1	36.9	36.9	36.6	36.4	35.1	32.9	30.6
		MCWB	24.0	24.6	25.7	26.8	27.8	28.7	28.0	28.5	28.8	27.6	26.0	24.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.6	27.6	28.9	30.3	31.0	30.6	30.7	30.7	30.0	28.2	26.7
		MCDB	30.3	30.8	32.3	34.3	35.9	36.4	36.2	36.1	36.2	35.2	33.4	31.2
	2%	WB	25.2	25.8	27.0	28.2	29.6	30.4	29.9	30.1	30.2	29.2	27.4	26.2
		MCDB	29.5	30.2	31.7	33.8	35.3	36.1	35.7	35.6	35.8	34.8	32.8	30.7
	5%	WB	24.8	25.2	26.5	27.8	29.2	30.0	29.4	29.6	29.8	28.8	27.0	25.8
		MCDB	29.1	29.6	31.2	33.4	34.9	35.7	35.3	35.2	35.4	34.5	32.3	30.3
	10%	WB	24.4	24.9	26.1	27.3	28.7	29.5	29.0	29.2	29.5	28.2	26.5	25.2
		MCDB	28.7	29.3	30.9	32.9	34.6	35.3	35.0	34.9	35.0	34.1	31.8	29.7
Mean Daily Temperature Range	5% DB	MDBR	6.8	6.6	6.7	7.1	7.5	6.4	5.9	6.5	7.3	8.2	7.5	7.1
		MCDBR	7.1	6.9	7.0	8.0	8.2	6.9	6.2	6.7	7.6	8.8	8.1	7.2
		MCWBR	4.0	3.7	4.0	4.7	4.7	4.5	4.2	4.1	4.5	4.7	4.1	3.9
	5% WB	MCDBR	6.8	6.7	6.5	7.1	7.2	6.2	5.9	6.4	7.2	8.2	7.3	6.8
		MCWBR	4.1	3.9	4.0	4.8	4.6	4.5	4.5	4.3	4.6	4.8	4.1	4.0
Clear-Sky Solar Irradiance	taub	0.500	0.554	0.588	0.591	0.649	0.733	0.795	0.775	0.682	0.566	0.529	0.497	
	taud	2.070	1.926	1.853	1.860	1.730	1.565	1.504	1.535	1.676	1.894	1.982	2.068	
	Ebn at Noon	784	761	750	749	697	633	597	615	671	741	749	771	
	Edh at Noon	158	191	211	210	237	276	294	287	248	194	171	155	
All-Sky Solar Radiation	RadAvg	4.93	5.65	6.45	6.99	7.27	7.04	6.44	6.37	6.56	6.22	5.28	4.90	
	RadStd	0.25	0.25	0.29	0.26	0.17	0.20	0.26	0.22	0.13	0.18	0.17	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.54	+0.64	N/A	+0.28	N/A	N/A	N/A	N/A	+201	+200
Regional (2 neighbors)	+0.53	+0.45	N/A	+0.50	N/A	N/A	N/A	-12	+200	+152

Nomenclature: See separate page